

Step 1 — Setting Up Environment

- Install Java 11

```
# apt install openjdk-11-jdk
```

- Check Java version

```
# java -version  
openjdk version "11.0.8" 2020-07-14
```

- Download Tomcat from source

```
# cd /opt  
  
# wget https://downloads.apache.org/tomcat/tomcat-9/v9.0.37/bin/apache-  
tomcat-9.0.37.tar.gz
```

- Extract the file

```
# tar -xvzf apache-tomcat-9.0.37.tar.gz  
  
# sudo mv apache-tomcat-9.0.37 tomcat
```

- Create Tomcat User & change ownership

```
# sudo useradd tomcat  
  
# sudo chown -R tomcat:tomcat tomcat  
  
# sudo chmod -R 755 tomcat
```

- Make the service run as a service

```
# sudo vi /opt/systemd/system/tomcat.service
```

[Unit]

```
Description=Tomcat 9 servlet container  
After=network.target
```

[Service]

```
Type=forking  
User=tomcat  
Group=tomcat  
Environment="JAVA_HOME=/usr/lib/jvm/java-11-openjdk-amd64"  
Environment="JAVA_OPTS=-Djava.security.egd=file:///dev/urandom -Djava.awt.headless=true"  
Environment="CATALINA_BASE=/etc/tomcat"  
Environment="CATALINA_HOME=/etc/tomcat"  
Environment="CATALINA_PID=/etc/tomcat/temp/tomcat.pid"  
Environment="CATALINA_OPTS=-Xms512M -Xmx1024M -server -XX:+UseParallelGC"  
ExecStart=/opt/tomcat/bin/startup.sh  
ExecStop=/opt/tomcat/bin/shutdown.sh
```

[Install]

```
WantedBy=multi-user.target
```

- Restart the daemon

```
# sudo systemctl daemon-reload
```

- Start the Tomcat

```
# sudo systemctl start tomcat
```

Step 2 — Create a Java Application

- Install a Java build tool

```
# sudo yum install maven -y
```

- Check Maven version

```
# mvn -version
```

Apache Maven 3.6.3

- Generate project

```
# mvn archetype:generate -DgroupId=com.app.example -DartifactId=java-app -  
DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```

The folder structure should be like this:

```
java-app/  
├── pom.xml  
└── src  
    ├── main  
    │   ├── java  
    │   │   ├── com  
    │   │   │   ├── app  
    │   │   │   │   ├── example  
    │   │   │   │   │   App.java  
    │   └── test  
    │       ├── java  
    │       │   ├── com  
    │       │   │   ├── app  
    │       │   │   │   ├── example  
    │       │   │   │   │   AppTest.java
```

- Build the application

```
# mvn package
```

- Copy application to Tomcat directory

```
# cp java-app/target/hello-world-1.war /opt/tomcat/webapps/
```

```
# systemctl restart tomcat
```

- Check the service